LFER Parameters

The following parameters are taken from Appendix 3 of "Free Energy Relationships in Organic and Bio-organic Chemistry" by Andrew Williams, The Royal Society of Chemistry, Cambridge, **2003**, pp 258-277.

APPENDIX 3

Tables of Structure Parameters

The structure parameters are collected from various sources as indicated. The present tables are meant to provide a convenient source for the reader rather than a critical assessment. Each value can vary depending on its source and if the constants are to be used for critical work they should be checked against the literature which is quoted in the footnotes. Exner[a] provides an excellent critical and extensive compilation of structure parameters.

[a] From O. Exner, *Correlation Analysis in Chemistry: Recent Advances*, N.B. Chapman and J. Shorter (eds.), Plenum Press, New York, 1978, p.439.

Table 1 *Hammett, Taft, Charton, σ_p^+, σ_p^-F, R and MR constants for some substituents*[b]

Formula[c]	*Structure*	*MR*[d]	σ_p^{+} [e]	σ_p^{-} [e]	*F*[f]	*R*[f]	σ_I [a]	σ_{meta} [g]	σ_{para} [g]	σ^* [g]
Br	−Br	8.88	0.15	0.25	0.45 *0.72*	−0.22 *−0.18*	0.45	0.39	0.23	2.84
Cl	−Cl	6.03	0.11	0.19	0.42	−0.19 *−0.24*	0.47	0.37	0.23	2.96
F	−F	0.92	−0.07	−0.03	0.45	−0.39	0.52	0.34	0.06	3.21
H	−H	1.03	0.00	0.00	0.00 *0.00*	0.00 *0.00*	0.00	0.00	0.00	0.49
I	−I	13.94	0.14	0.27	0.42 *0.65*	−0.24 *−0.12*	0.39	0.35	0.18	2.46
N_3	$-N_3$	10.20		0.11	0.48	−0.40	0.42	0.27	0.15	2.62
O	$-O^-$		−2.30	−0.82	−0.26	−0.55	−0.16	−0.47	−0.81	−2.78
S	$-S^-$		−2.62		0.03	−1.24		−0.36		
BF_2	$-BF_2$				0.26	0.22	0.16	0.32	0.48	
Br_3Ge	$-GeBr_3$				0.61	0.12	0.59	0.66	0.73	3.7
Br_3Si	$-SiBr_3$				0.44	0.13	0.39	0.48	0.57	2.4
Cl_3Ge	$-GeCl_3$		0.57		0.65	0.14	0.63	0.71	0.79	3.9
Cl_3Si	$-SiCl_3$		0.57		0.44	0.12	0.39	0.48	0.56	2.4
F_3Ge	$-GeF_3$				0.76	0.21	0.74	0.85	0.97	4.6
F_3S	$-SF_3$				0.63	0.17	0.60	0.70	0.80	3.75
F_3Si	$-SiF_3$				0.47	0.22	0.42	0.54	0.66	2.62
GeH_3	$-GeH_3$				0.03	−0.02		0.00	0.01	~0.7
HO	−OH	2.85	−0.92	−0.37	0.33 *0.46*	−0.70 *−1.89*	0.24	0.12	−0.37	1.34
HS	−SH	9.22	−0.03		0.30 *0.52*	−0.15 *−0.26*	0.26	0.25	0.15	1.68
H_2N	$-NH_2$	5.42	−1.30	−0.15	0.08 *0.38*	−0.74 *−2.52*	0.19	−0.16	−0.66	0.62

(*continued overleaf*)

Table 1 (*continued*)

Formula[c]	Structure	MR[d]	σ_p^{+}[e]	σ_p^{-}[e]	F[f]	R[f]	σ_I[a]	σ_{meta}[g]	σ_{para}[g]	σ^{*}[g]
H_3N	$-NH_3^+$			–0.56	0.92	–0.32	0.60	0.67	0.53	3.76
H_3N_2	$-NHNH_2$	8.44			0.22	–0.77	0.14	–0.02	–0.55	0.40
H_3Si	$-SiH_3$		0.14		0.06	0.04	0.09	0.05	0.10	
IO_2	$-IO_2$	63.51			0.61	0.17		0.70	0.76	3.71
					0.99	*0.99*				
NO	$-NO$	5.20		1.63	0.49	0.42	0.33			2.08
NO_2	$-NO_2$	7.36		1.24	0.65	0.13	0.67	0.72	0.78	4.25
					1.00	*1.00*				
NO_3	$-ONO_2$				0.48	0.22	0.62	0.55	0.70	3.86
O_2S	$-SO_2^-$			0.08	0.03	–0.08		–0.02	–0.05	
O_3P	$-PO_3^{2-}$			–0.16				–0.02		
O_3S	$-SO_3^-$			0.58	0.29	0.06	0.13	0.05	0.09	0.81
					–0.05	*0.53*				
$AsHO_3$	$-AsO_3H$			0.46	0.04	–0.06	0.01	0.00	–0.02	0.14
BH_2O_2	$-B(OH)_2$	11.04		0.45	–0.03	0.15	–0.08	–0.01	0.12	0.95
$ClOS$	$-SOCl$						0.68	0.75	0.82	4.25
ClO_2S	$-SO_2Cl$		0.38		1.16	–0.05	0.80	0.92	1.04	5.0
Cl_2OP	$-POCl_2$		0.33		0.70	0.20	0.65	0.78	0.90	4.1
Cl_2PS	$-PSCl_2$				0.63	0.17	0.59	0.70	0.80	3.7
FOS	$-SOF$			1.54	0.67	0.16	0.66	0.74	0.83	4.12
FO_2S	$-SO_2F$	8.65			0.72	0.19	0.75	~0.89	~0.99	~4.7
F_2OP	$-POF_2$				0.74	0.15		0.81	0.89	
HO_2P	$-PO(OH)$								0.14	
HO_2S	$-SO(OH)$				0.01	–0.08		–0.04	–0.07	
HO_3P	$-PO_3H$				0.19	0.07		0.25	0.17	1.41
					–0.19	*–0.32*				
HO_3S	$-SO_2OH$							0.55		
HO_4P	$-OPO_3H$				0.41	–0.41		0.29	0.00	
H_2NO	$-NHOH$	7.22			0.11	–0.45		–0.04	–0.34	0.30
H_2O_3P	$-PO(OH)_2$							0.36	0.42	
H_3NO	$-ONH_3^+$						0.47	0.53		2.92
H_2NO_2S	$-SO_2NH_2$	12.28		0.94	0.49	0.21	0.44	0.46	0.57	2.61
					0.55	*1.07*				
N_2	$-N_2^+$				1.58	0.33		1.76	1.91	
					2.36	*2.81*				
CBr_3	$-CBr_3$				0.28	0.01	0.26	0.28	0.29	2.43
CCl_3	$-CCl_3$				0.38	0.09	0.41	0.40	0.46	2.65
CF_3	$-CF_3$	5.02		0.65	0.38	0.16	0.42	0.46	0.53	2.61
					0.64	*0.76*				
CH_3	$-CH_3$	5.65	–0.31	–0.17	0.01	–0.18	–0.10	–0.06	–0.14	0.00
					–0.01	*–0.41*				
CN	$-CN$	6.33		0.88	0.51	0.15	0.58	0.56	0.66	3.30
					0.90	*0.71*				
	$-NC$				0.47	0.02	0.67	0.48	0.49	
CN_7	5-azido-1-tetrazolyl[h]				0.53	0.01		0.54	0.54	
CO_2	$-COO^-$	6.05	–0.02	0.31	–0.10	0.10	–0.17	–0.10	0.00	–1.06
					–0.27	*0.40*				
$CClN_4$	5-Cl-1-tetrazolyl[h]	23.60		0.70	0.58	0.03		0.80	0.61	
$CClO$	$-COCl$			1.24	0.46	0.15	0.38	0.53	0.69	2.37
CCl_3O	$-OCCl_3$				0.46	–0.11	0.51	0.43	0.35	3.19
CFO	$-COF$				0.48	0,22	0.39	0.55	0.70	2.44
CF_3O	$-OCF_3$	7.86		0.27	0.39	–0.04	0.55	0.36	0.33	
CF_3S	$-SCF_3$	13.81		0.64	0.36	0.14	0.44	0.38	0.50	2.75
$CHBr_2$	$-CHBr_2$				0.31	0.01	0.30	0.31	0.32	1.96
$CHCl_2$	$-CHCl_2$				0.312	0.01	0.31	0.31	0.32	1.94
CHF_2	$-CHF_2$				0.29	0.03	0.32	0.32	0.35	2.05
CHI_2	$-CHI_2$				0.27	–0.01	0.26	0.26	0.26	1.62
CHN_2	$-NHCN$	10.14		0.28	–0.22	0.37	0.21	0.06		
CHN_4	1-tetrazolyl[h]	18.33		0.57	0.52	–0.02		0.60	0.57	3.4

(*continued overleaf*)

Table 1 (*continued*)

Formula[c]	*Structure*	*MR*[d]	σ_p^{+} [e]	σ_p^{-} [e]	*F*[f]	*R*[f]	σ_I[a]	σ_{meta}[g]	σ_{para}[g]	σ^*[g]
CHO	–CHO	6.88		1.03	0.33	0.09	0.25	0.36	0.44	2.15
CHO_2	–COOH	6.93		0.77	0.34	0.11	0.39	0.35	0.44	2.08
					0.44	*0.66*				
CH_2Br	$-CH_2Br$	13.39	0.02		0.14	0.00	0.18	0.12	0.14	1.00
CH_2Cl	$-CH_2Cl$	10.49	–0.01		0.13	–0.01	0.16	0.11	0.12	1.05
CH_2F	$-CH_2F$				0.15	–0.04	0.18	0.11	0.10	1.10
CH_2I	$-CH_2I$	18.60			0.12	–0.01	0.16	0.10	0.11	0.85
CH_3O	$-CH_2OH$	7.19	–0.04	0.08	0.03	–0.03	0.11	0.01	0.01	0.54
					0.54	*–1.68*				
	$-OCH_3$.	7.87	–0.78	–0.26	0.29	–0.56	0.30	0.11	–0.27	1.81
CH_3O_2	$-CH(OH)_2$						0.22			1.37
CH_3S	$-CH_2SH$						0.10	0.08		0.62
	$-SCH_3$	13.82	–0.60	0.06	0.23	–0.23	0.25	0.15	0.00	1.56
					0.68	*–1.30*				
CH_3S_2	$-SSCH_3$				0.27	–0.14		0.22	0.13	
CH_3Se	$-SeCH_3$	17.03			0.16	–0.16		0.1	0.00	0.95
					0.28	*–0.39*				
CH_4N	$-CH_2NH_2$				0.04	–0.15	0.08	–0.03	–0.11	0.50
	$-NHCH_3$	10.33	–1.81		0.03	–0.73	0.18	–0.30	–0.84	–0.81
CH_5N	$-CH_2NH_3^+$				0.59	–0.06	–0.36	0.32	0.29	2.24
	$-NH_2CH_3^+$						0.60	0.96		3.74
CNO	–OCN				0.69	–0.15	0.80	0.67	0.54	5.0
	–NCO		–0.19		0.31	–0.12	0.36	0.27	0.19	2.25
CNS	–NCS	17.24		0.34	0.51	–0.13	0.42	0.48	0.38	2.62
	–SCN	13.40		0.59	0.49	0.03	0.58	0.41	0.52	3.43
CN_3O_6	$-C(NO_2)_3$				0.65	0.17	0.73	0.72	0.82	4.6
CHN_4O	5-OH-1-tetrazolyl[h]				0.41	–0.08		0.39	0.33	
CHN_4S	5-SH-1-tetrazolyl[h]				0.44	–0.01		0.45	0.45	
	1,2,3,4-Thia-triazol-5-yl-amino[i]				0.35	–0.16		0.30	0.19	2.62
CH_2ClO	$-OCH_2Cl$				0.33	–0.25	0.41	0.25	0.08	2.56
CH_2FO	$-OCH_2F$				0.29	–0.27	0.37	0.20	0.02	2.31
CH_2FS	$-SCH_2F$				0.25	–0.05	0.27	0.23	0.20	1.69
CH_2NO	$-CONH_2$	9.81		0.61	0.26	0.10	0.27	0.28	0.31	1.68
	–NHCHO	10.31			0.28	–0.28	0.32	0.19	0.00	1.62
	–CH=NOH	10.28			0.28	–0.18		0.22	0.10	
CH_3Br_2Si	$-SiBr_2(CH_3)$								0.29	
CH_3Cl_2Si	$-SiCl_2(CH_3)$		0.08		0.29	0.10	0.24	0.31	0.39	1.5
CH_3F_2Si	$-SiF_2(CH_3)$		0.23		0.32	–0.09		0.29	0.23	
CH_3N_2O	$-NHCONH_2$	13.72			0.09	–0.33	0.21	–0.03	–0.24	1.31
CH_3N_2S	$-NHCSNH_2$	22.19			0.26	–0.1	0.29	0.22	0.16	1.8
CH_3OS	$-SOCH_3$				0.24	–0.07	0.25	0.21	0.17	1.56
	$-S(O)CH_3$	13.70		0.73	0.52	–0.03	0.50	0.52	0.49	2.88
					0.80	*0.45*				
CH_3O_2S	$-SO_2CH_3$	13.49		1.13	0.53	0.19	0.59	0.64	0.73	3.68
					0.80	*0.85*				
	$-S(O)OCH_3$			0.89	0.47	0.07	0.45	0.50	0.54	2.84
	$-OS(O)CH_3$				0.43	0.02		0.44	0.45	
CH_3O_3S	$-OSO_2CH_3$	16.99	0.16		0.40	–0.04		0.39	0.36	3.62
CH_2NOS	$-SCONH_2$						0.33	0.34		2.07
CH_4NOS	$-NHSO_2CH_3$	18.17			0.28	–0.25	0.42	0.20	0.03	
CHF_3NO_2S	$-NHSO_2CF_3$						0.49	0.44	0.39	3.1
C_2F_5	$-CF_2CF_3$	9.23		0.69			0.41	0.50	0.52	2.56
C_2H	$-C\equiv CH$	9.55	0.18	0.53	0.22	0.01	0.35	0.20	0.23	2.18
C_2H_3	$-CH=CH_2$	10.99	–0.16		0.13	–0.17	0.09	0.08	–0.08	0.56
C_2H_5	$-C_2H_5$	10.30		0.69	0.00	–0.15	–0.05	–0.07	–0.15	–0.10
					–0.02	*–0.44*				
C_2F_3O	$-COCF_3$			1.09			0.47	0.63	0.80	3.7

(*continued overleaf*)

Table 1 (*continued*)

Formula[c]	*Structure*	*MR*[d]	$\sigma_p^{+\,e}$	$\sigma_p^{-\,e}$	F^f	R^f	σ_I^a	σ_{meta}^g	σ_{para}^g	σ^{*g}
C_2F_5O	$-OCF_2CF_3$			0.61				0.48	0.28	
$C_2F_3O_2$	$-OCOCF_3$						0.65	0.56	0.46	4.06
C_2HCl_2	$-CH{=}CCl_2$						0.16	0.11		1.00
C_2HS	$-SC{\equiv}CH$						0.32	0.26	0.19	2.00
$C_2H_2Cl_3$	$-CH_2CCl_3$						0.12	0.06		0.75
$C_2H_2F_3$	$-CH_2CF_3$			0.14	0.15	−0.06	0.14	0.16	0.14	0.92
C_2H_2N	$-CH_2CN$	10.11	0.16	0.11	0.17	0.01	0.18	0.16	0.18	1.30
$C_2H_2O_2$	$-CH_2COO^-$		−0.53	−0.16			0.01			−0.06
C_2H_3O	$-COCH_3$	11.18		0.84	0.33	0.17	0.29	0.38	0.50	1.65
					0.50	*0.90*				
$C_2H_3O_2$	$-OCOCH_3$	12.47	−0.19		0.42	−0.11	0.41	0.39	0.31	2.56
					0.70	*−0.04*				
	$-CH_2COOH$		−0.01	0.05			0.17		−0.07	1.05
	$-COOCH_3$	12.87		0.75	0.34	0.11	0.34	0.32	0.39	2.00
$C_2H_3S_2$	$-SC(S)CH_3$						0.48			3.00
C_2H_4Cl	$-CH_2CH_2Cl$									0.385
C_2H_5O	$-OC_2H_5$	12.47	−0.81	−0.28	0.26	−0.50	0.27	0.10	−0.24	1.68
					0.61	*−1.72*				
	$-CH_2OCH_3$	12.07	−0.05		0.13	−0.12	0.11	−0.10	0.03	0.52
	$-CH_2CH_2OH$						0.060			
C_2H_5S	$-SC_2H_5$	18.42			0.26	−0.23	0.25	0.23	0.03	1.56
C_2H_6N	$-N(CH_3)_2$	15.55	−1.70	−0.12	0.15	−0.98	0.10	−0.15	−0.83	0.32
					0.69	*−3.81*				
	$-NHC_2H_5$	14.98			−0.04	−0.57		−0.24	−0.61	−0.62
C_2H_6S	$-S(CH_3)_2^+$			0.83	0.98	−0.08	0.90	1.00	0.90	5.09
					1.62	*0.52*				
C_2H_7N	$-CH_2CH_2NH_3^+$							0.23	0.17	
	$-NH(CH_3)_2^+$						0.70	0.84		4.36
	$-NH_2C_2H_5^+$							0.96		3.74
C_2F_3OS	$-SCOCF_3$						0.51	0.48	0.46	3.19
C_2H_4Cl	$-CH_2CH_2Cl$									0.385
C_2H_3OS	$-SCOCH_3$	18.42		0.46	0.37	0.07	0.41	0.39	0.44	2.29
					0.53	*0.68*				
C_2H_4NO	$-NHCOCH_3$	14.93	−0.60	−0.46	0.31	−0.31	0.28	0.21	0.00	1.40
					0.77	*−1.43*				
	$-CH_2CONH_2$				0.08	−0.01	0.05	0.06	0.07	0.31
	$-CONHCH_3$	14.57			0.35	−0.01	0.29	0.35	0.36	
$C_2H_4NO_2$	$-CH_2CH_2NO_2$									0.50
C_2H_4NS	$-CSNHCH_3$				0.29	0.05		0.30	0.34	
	$-NHCSCH_3$	23.40			0.30	−0.18		0.24	0.12	
$C_2H_5O_2S$	$-SO_2C_2H_5$	18.14			0.59	0.18	0.60	0.66	0.77	3.74
	$-CH_2SO_2CH_3$							0.18		1.32
C_2H_6OP	$-PO(CH_3)_2$			0.72	0.40	0.10	0.35	0.43	0.50	2.81
$C_2H_6O_3P$	$-PO(OCH_3)_2$	21.87			0.37	0.16	0.24	0.42	0.55	
$C_2H_6O_4P$	$-OPO(OCH_3)_2$								0.04	
C_2H_3ClNO	$-NHCOCH_2Cl$	19.77			0.27	−0.30	0.36	0.17	−0.03	2.06
C_2H_6ClNP	$-P(Cl)N(CH_3)_2$				0.31	0.25		0.38	0.56	
$C_2H_6NO_2S$	$-SO_2N(CH_3)_2$		0.86	0.88	0.44	0.21	0.42	0.51	0.65	2.62
C_3F_7	$-CF(CF_3)_2$			0.68			0.48	0.37	0.53	3.00
C_3H_3	$-C{\equiv}CCH_3$				0.29	−0.26	0.21	0.10	0.12	1.20
	$-CH_2C{\equiv}CH$						0.13	0.07		0.81
C_3H_5	$-cyclo\text{-}C_3H_5$	13.53	−0.41	−0.09	0.02	−0.23		−0.07	−0.21	
	$-CH{=}CHCH_3$									0.36
C_3H_7	$-CH(CH_3)CH_3$	14.96	−0.28	−0.16	0.04	−0.19	−0.02	−0.07	−0.15	−0.19
	$-CH_2CH_2CH_3$	14.96	−0.29	−0.06	0.01	−0.14	−0.02	−0.05	−0.15	−0.115
C_3HF_6	$-CH(CF_3)_2$						0.21			1.32
C_3H_3O	$-CH{=}CHCHO$				0.29	−0.16		0.24	0.13	
$C_3H_3O_2$	$-CH{=}CHCOOH$							0.14	0.90	1.00
$C_3H_4F_3$	$-CH_2CH_2CF_3$									0.32

(*continued overleaf*)

Table 1 (*continued*)

Formula[c]	*Structure*	*MR*[d]	σ_p^{+} [e]	σ_p^{-} [e]	*F*[f]	*R*[f]	σ_I[a]	σ_{meta}[g]	σ_{para}[g]	σ^*[g]
C_3H_4O	$-CH_2CH_2COO^-$									0.02
$C_3H_5O_2$	$-CH_2OCOCH_3$				0.07	−0.02	0.16	0.04	0.05	1.00
	$-CH_2CH_2COOH$	16.52			0.02	−0.09		−0.03	−0.07	0.35
	$-COOC_2H_5$	17.47		0.64	0.34	0.11	0.34	0.37	0.45	2.26
					0.47	*0.67*				
	$-CH_2COOCH_3$			0.07			0.17	0.13		1.06
C_3H_7O	$-CH_2OC_2H_5$	16.72								0.58
	$-CH_2CH(OH)\text{-}CH_3$				−0.06	−0.11	−0.01	−0.12		−0.06
	$-OCH(CH_3)CH_3$	17.06	−0.85		0.34	−0.79	0.26	0.05	−0.45	1.62
					0.90	*−2.88*				
	$-OCH_2CH_2CH_3$	17.06	−0.83		0.26	−0.51	0.27	0.10	−0.25	1.68
					0.63	*−1.77*				
C_3H_7S	$-SCH(CH_3)CH_3$				0.30	−0.23	0.25	0.23	0.07	1.56
	$-SCH_2CH_2CH_3$	23.07					0.24	0.22		1.49
C_3H_9Ge	$-Ge(CH_3)_3$				0.03	−0.03	0.00	0.00	0.00	
C_3H_9N	$-CH_2NH(CH_3)^+$		0.50		0.39	0.04		0.40	0.43	
	$-NH_2CH_2CH_2\text{-}CH_3^+$						0.60	0.71		3.74
	$-N(CH_3)_3^+$	21.20	0.41	0.77	0.86	−0.04	0.73	0.88	0.82	4.55
					1.54	*(0.00)*				
C_3H_9P	$-P(CH_3)_3^+$				0.71	0.02	0.40	0.50	0.80	2.5
C_3H_9Si	$-Si(CH_3)_3$	24.96	0.02		0.01	−0.08	−0.13	−0.04	−0.07	−0.81
					−0.10	*0.16*				
C_3H_9Sn	$-Sn(CH_3)_3$		−0.12		0.03	−0.03	0.00	0.0	0.0	
C_3H_6NO	$-CH_2NHCOCH_3$						0.07	−0.04	−0.05	0.43
	$-NHCOC_2H_5$	21.18						0.23		1.56
$C_3H_6NO_2$	$-OCON(CH_3)_2$									2.87
	$-NHCOOC_2H_5$						0.26	0.07	−0.15	1.99
$C_3H_7N_2O$	$-NHCONHC_2H_5$							0.04	−0.26	
$C_3H_7N_2S$	$-NHCSNHC_2H_5$	31.66						0.30	0.07	
C_4F_9	$-C(CF_3)_3$			0.71			0.35	0.35	0.52	
$C_4H_6F_3$	$-CH_2CH_2CH_2CF_3$									0.12
C_4H_7	$-CH{=}CHC_2H_5$						0.05	−0.04		0.31
	$-CH_2CH{=}CHCH_3$						0.00	−0.04		0.13
	$-CH{=}C(CH_3)_2$						0.03	−0.06		0.19
C_4H_9	$-(CH_2)_3CH_3$	19.61	−0.29	−0.12	−0.01	−0.15	−0.04	−0.07	−0.16	−0.13
	$-CH(CH_3)C_2H_5$				−0.02	−0.10	−0.03	−0.08	−0.19	−0.125
	$-C(CH_3)_3$	19.62	−0.26	−0.13	−0.02	−0.18	−0.07	−0.10	−0.20	−0.30
					−0.11	*−0.29*				
	$-CH_2CH(CH_3)CH_3$			0.01			−0.03	−0.7	−0.12	−0.21
C_4N_3	$-C(CN)_3$				0.92	0.04	0.98	0.98	0.99	6.1
$C_4H_2F_7$	$-CH_2C_3F_7$						0.14	0.08	0.87	
C_4H_3O	2-furyl		−0.39	0.21	0.1	−0.08	0.17	0.06	0.02	0.25
C_4H_3S	2-thienyl	24.04	−0.43	0.13	0.13	−0.08	0.21	0.09	0.05	1.31
	3-thienyl	24.04			0.08	−0.10		0.03	−0.02	0.62
C_4H_4N	1-pyrrolyl	21.85			0.50	−0.13		0.47	0.37	
C_4H_9O	$-O(CH_2)_3CH_3$	21.66			0.29	−0.61	0.27	−0.05	−0.32	1.68
					0.72	*−2.16*				
	$-OCH(CH_3)C_2H_5$						0.26	0.25		1.62
	$-CH_2C(OH)(CH_3)_2$				−0.11	−0.06	−0.04	−0.16		−0.25
$C_4H_9O_3$	$-C(OCH_3)_3$				0.01	−0.05	−0.03	−0.03	−0.04	
C_4H_9S	$-S(CH_2)_3CH_3$						0.23	0.21		1.44
	$-SCH(CH_3)C_2H_5$									1.32
$C_4H_{10}N$	$-NH(CH_2)_3CH_3$	24.26			−0.21	−0.30		−0.34	−0.51	−1.08
	$-N(C_2H_5)_2$	24.85	−2.07	−0.43	0.01	−0.73	0.16	−0.15	−0.53	
$C_4H_{11}N$	$-NH_2(CH_2)_3CH_3^+$						0.60	0.71		3.74
	$-NH_2C(CH_3)_3^+$							0.71		

(continued overleaf)

Table 1 (*continued*)

Formula[c]	*Structure*	*MR*[d]	σ_p^{+} [e]	σ_p^{-} [e]	*F*[f]	*R*[f]	σ_I[a]	σ_{meta}[g]	σ_{para}[g]	σ^*[g]
	$-CH_2CH_2NH(CH_3)_2^+$				0.29	−0.15		0.24	0.14	
	$-NH_2CH_2CH(CH_3)_2^+$						0.60			3.60
	$-CH_2N(CH_3)_3^+$		0.50		0.38	0.06	0.30	0.40	0.44	1.90
$C_4H_{11}Si$	$-CH_2Si(CH_3)_3$	29.61	−0.66		−0.09	−0.12	−0.05	−0.16	−0.21	−0.26
					−0.19	*−0.32*				
C_4H_8NO	−morpholino[j]									0.69
$C_4H_8NO_2$	$-NHCH_2COOC_2H_5$							−0.10	−0.68	
$C_4H_{10}O_3P$	$-P(O)(OC_2H_5)_2$	31.16	0.54	0.84			0.35	0.49	0.57	3.02
$C_4H_{10}NO_2$	$-NH(CH_2CH_2OH)_2^+$									4.43
C_5H_{11}	$-(CH_2)_2CH(CH_3)_2$								−0.23	
	$-CH_2C(CH_3)_3$		−0.31		0.03	−0.14	−0.02	−0.13	−0.12	−0.165
	$-(CH_2)_4CH_3$	24.26		−0.19	−0.01	−0.14	−0.04	−0.08	−0.15	−0.23
	$-C(CH_3)_2C_2H_5$								−0.19	
	$-CH(C_2H_5)_2$									−0.225
C_5H_4N	2-pyridyl			0.55	0.40	−0.23	0.20	0.33	0.17	
C_5H_5S	2-thenyl[k]						0.05	−0.04		0.31
C_5H_9O	$-COC(CH_3)_3$				0.26	0.06		0.27	0.33	
$C_5H_{10}N$	−piperidino[l]								−0.12	
$C_5H_{13}N$	$-CH_2CH_2NMe_3^+$				0.19	−0.06		0.16	−0.01	
C_6C_{15}	$-C_6C_{15}$				0.27	−0.03	0.25	0.24	0.24	1.56
C_6F_5	$-C_6F_5$			0.43	0.27	0.00	0.31	−0.12	−0.03	
C_6H_5	$-C_6H_5$	25.36	−0.18	0.02	0.12	−0.13	0.12	0.06	−0.01	0.60
					0.55	*1.07*				
C_6H_{11}	−cyclohexyl		−0.29	−0.14	0.03	−0.18	−0.02	−0.14	−0.22	−0.15
C_6H_{13}	$-(CH_2)_5CH_3$						−0.06	−0.16		−0.25
	$-CH(CH_3)C(CH_3)_3$									−0.28
$C_6H_5N_2$	$-N{=}N\text{-}C_6H_5$	31.31	−0.19	0.45	0.30	0.09	0.19	0.29	0.31	1.87
C_6H_5O	$-OC_6H_5$	27.68	−0.50	−0.10	0.37	−0.40	0.34	0.25	−0.32	2.43
					0.76	*−1.29*				
C_6H_5S	$-SC_6H_5$		−0.55	0.18	0.30	−0.23	0.30	0.17	0.13	1.87
C_6H_5Se	$-SeC_6H_5$		−0.47	−0.29			0.37		0.13	2.30
C_6H_6N	$-NHC_6H_5$	30.04		0.76	0.22	−0.78	–	−0.12	−0.45	
C_6H_5OS	$-S(O)C_6H_5$			1.21	0.51	−0.07	0.52	0.51	0.46	3.24
$C_6H_5O_2S$	$-SO_2C_6H_5$	33.20		1.11	0.58	−0.10	0.57	0.62	0.70	3.25
$C_6H_5O_3S$	$-S(O)_2OC_6H_5$						0.55		0.51	
	$-OSO_2C_6H_5$	36.70			0.37	−0.04	0.58	0.36	0.33	3.62
$C_6H_{14}O_3P$	$-P(O)(OC_3H_7)_2$				0.33	0.17		0.38	0.50	
$C_6H_{15}O_3Si$	$-Si(OC_2H_5)_3$		0.17		0.03	0.05		0.02	0.08	
$C_6H_6NO_2S$	$-SO_2NHC_6H_5$				0.51	0.14		0.56	0.65	
	$-NHSO_2C_6H_5$	37.88	−0.98		0.24	−0.23	0.32	0.16	0.01	1.99
C_7H_7	$-CH_2C_6H_5$	30.01	−0.28	−0.09	−0.04	−0.05	0.04	−0.08	−0.09	0.215
C_7H_5O	$-COC_6H_5$	30.33		0.83	0.31	0.12	0.27	0.36	0.46	2.2
$C_7H_5O_2$	$-OCOC_6Hi$	32.33	−0.07		0.26	−0.13		0.21	0.13	2.57
	$-COOC_6H_5$	32.31			0.34	0.10		0.37	0.44	
C_7H_6N	$-N{=}CHC_6H_5$	33.01			0.14	−0.69		−0.08	−0.55	
	$-CH{=}NC_6H_5$	33.01			0.33	0.09		0.35	0.42	
C_7H_7O	$-OCH_2C_6H_5$								−0.41	
	$-CH(OH)C_6H_5$				0.05	−0.08	0.08	0.00	−0.03	0.50
	$-CH_2OC_6H_5$	32.19			0.08	−0.01	0.16	0.04	0.05	0.85
C_7H_7S	$-SCH_2C_6H_5$						0.25	0.23		1.56
C_7H_6NO	$-CONHC_6H_5$						0.27	0.23	0.41	1.56
	$-NHCOC_6H_5$	34.64	−0.60				0.25	0.22	0.08	1.68
C_7H_{13}	$-CH_2\mathit{cyclo}C_6H_{11}$									−0.06
C_9H_{11}	$-CH(C_2H_5)C_6H_5$									0.04
	$-CH_2CH_2CH_2C_6H_5$									0.02
C_8H_5	$-C{\equiv}CC_6H_5$	33.21	−0.03	0.30	0.15	0.01	0.22	0.14	0.16	1.35
C_8H_7	$-CH{=}CHC_6H_5$	34.17	−1.00	0.13	0.10	−0.17	0.07	0.03	−0.07	0.41

(*continued overleaf*)

Table 1 (*continued*)

Formula[c]	*Structure*	*MR*[d]	σ_p^{+} [e]	σ_p^{-} [e]	*F*[f]	*R*[f]	σ_I[a]	σ_{meta}[g]	σ_{para}[g]	σ^*[g]
C_8H_9	$-CH_2CH_2C_6H_5$		−0.28	−0.12	−0.01	−0.11	−0.01	−0.07	−0.12	0.08
	$-CH(CH_3)C_6H_5$						0.06	−0.03		0.11
C_8H_6N	3-indolyl						0.01	−0.12		−0.06
$C_{10}H_7$	(α-naphthyl						0.12	0.06		0.75
	β-naphthyl						0.12	0.06		0.75
$C_{10}H_{15}$	–adamantyl[m]		−0.27	−0.14	−0.07	−0.06		−0.12	−0.24	
$C_{12}H_{10}OP$	$-PO(C_6H_5)_2$	59.29	0.49	0.68	0.32	0.21	0.27	0.40	0.54	1.71
$C_{12}H_{10}PS$	$-PS(C_6H_5)_2$			0.73	0.23	0.24	0.40	0.29	0.47	
$C_{13}H_{11}$	$-CH(C_6H_5)_2$		−0.19		0.01	−0.06	0.07	−0.03	−0.04	0.405
$C_{18}H_{15}Si$	$-Si(C_6H_5)_3$		0.12	0.29	−0.04	0.14	0.13	−0.03	0.10	
$C_{19}H_{15}$	$-C(C_6H_5)_3$		−0.21		0.01	0.01	0.10	−0.01	0.02	−0.4
$C_{19}H_{15}S$	$-SC(C_6H_5)_3$						0.12	0.05		0.69
C_n	general									
OR	–OR		−0.83				0.27	0.1	−0.35	1.69
SR	–SR						0.25	0.18	0.05	1.56
HNR	–NHR						0.17	−0.29	−0.56	
H_2NR	$-NH_2R^+$						0.60			3.75
CO_2R	–COOR			0.74			0.31	0.35	0.44	1.94
O_2PR_2	$-P(OR)_2$						0.09	0.12	0.15	
O_2SR	$-SO_2R$						0.57	0.64	0.73	3.56
O_3PR_2	$-PO(OR)_2$						0.28	0.38	0.50	
O_3SR	$-SO_2OR$						0.50	0.71	0.90	3.12
CHO_2R_2	$-CH(OR)_2$						−0.02	−0.04	−0.05	1.12
CHS_2R_2	$-CH(SR)_2$						0.15			0.94
CH_2OR	$-CH_2OR$						0.10	0.02	0.02	
$C_2H_2O_2R$	$-CH_2COOR$						0.17			1.12
$C_2H_2O_3R$	$-OCH_2COOR$						–		−0.18	
$C_3H_2O_2R$	–CH=CH-COOR						0.18	0.19	0.03	1.12
$CHNO_2R$	–NHCOOR						0.29	0.07	−0.15	
CH_2O_2SR	$-CH_2SO_2R$				0.16	0.01	0.21	0.15	0.17	1.12
CH_3NO_2R	$-NHCH_2COOR$						0.26	−0.10	−0.68	

Appendix 3

[a] σ_I from O. Exner, *Correlation Analysis in Chemistry: Recent Advances*, N.B. Chapman and J. Shorter (eds.), Plenum Press, New York, 1978, p 439.

[b] Parameters obtained mainly from footnotes a to d; other databases which can be consulted for parameters are:
F.E. Norrington *et al.*, *J. Med. Chem.*, 1975, **18**, 604; D.H. McDaniel and H.C. Brown, *J. Org. Chem.*, 1958, **23**, 420; M. Charton, *Progr. Phys. Org. Chem.*, 1981, **13**, 119; M. Charton, *J. Org. Chem.*, 1964, **29**, 1222; H.H. Jaffé, *Chem. Rev.*, 1953, **53**, 191.

[c] The molecular formula sequence is that adopted by Perrin (footnote g) where the formulae are blocked together initially by their carbon number and then by the number of elements.

[d] MR (Molar refractive index) from C. Hansch and A. Leo, *Substituent Constants for Correlation Analysis in Chemistry and Biology*, John Wiley, New York, 1979; C. Hansch *et al.*, *J. Med. Chem.*, 1973, **16**, 1207.

[e] σ_p^+ and σ_p^- from C. Hansch *et al.*, *Chem. Rev.*, 1991, **91**, 165.

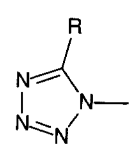
[f] *F* and *R* parameters are those from C. Hansch *et al.*, *Chem. Rev.*, 1991, **91**, 165; they are defined from different databases than are those in Swain's papers. Hansch's *F* and *R* are derived with the following equations:

$$F = \sigma_I = 1.297.\sigma_m - 0.385.\sigma_p + 0.033$$

$$R = \sigma_p - F$$

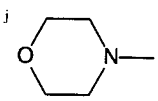
The quoted *F* and *R* parameters were derived by Hansch from σ_m, σ_p and σ_I values different from those given here. The *italicised F* and *R* parameters are those defined by C.G. Swain, *J. Am. Chem. Soc.*, 1983, **105**, 492.

[g] σ_m, σ_p, and σ^* from D.D. Perrin *et al.*, *pK$_a$ Prediction for Organic Acids and Bases*, Chapman & Hall, London, 1981.



R = N_3, OH, Cl, H, SH

j



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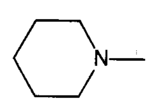

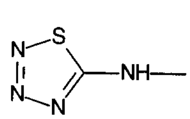

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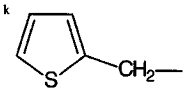

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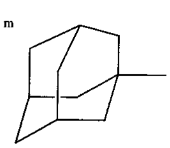

Table 2 *Charton's v_x (corrected atomic radii) and Taft–Pavelich steric parameters*[a]

Substituent	E_S	v_x	*Substituent*	E_S	v_x
Me	(0.00)	0.52	Bu_2CH		1.56
i-Bu	−0.93	0.98	*n*-C_5H_{11}	−0.40	0.68
t-Bu	−1.54	1.24	$(i\text{-}PrCH_2)_2CH$		1.70
CCl_3	−2.06	1.38	$(t\text{-}BuCH_2)_2CH$	−3.18	2.03
CBr_3	−2.43	1.56	*i*-PrCHEt		2.11
CI_3	−0.90	1.79	*t*-BuCHMe	−3.33	2.11
CF_3	−1.16	0.91	*t*-$BuCH_2CHMe$	−1.85	1.41
Me_3Si		1.40	H-	1.24	(0.00)
F		0.27	*t*-$BuCMe_2$	−3.90	2.43
Cl		0.55	*t*-$BuCH_2CMe_2$	−2.57	1.74
Br		0.65	Et_3C	−3.80	2.38
I		0.78	$PhCH_2$	−0.38	0.70
Ph	−2.48	1.66	$PhCH_2CH_2$	−0.38	0.70
$Ph(CH_2)_3$	−0.45	0.70	$CHPh_2$	−1.43	1.25
Et	−0.07	0.56	$Ph(CH_2)_4$	−1.76	0.70
Pr	−0.36	0.68	PhMeCH	−1.19	0.99
n-Bu	−0.39	0.68	PhEtCH	−1.50	1.18
$BuCH_2$	−0.40	0.68	Ph_2CH	−1.43	1.25
$BuCH_2CH_2$	−0.30	0.73	$ClCH_2$	−0.24	0.60
$Bu(CH_2)_3$		0.73	$BrCH_2$	−0.27	0.64
$Bu(CH_2)_4$	−0.33	0.68	Cl_2CH	−1.54	0.81
i-$PrCH_2$	−0.93	0.98	CH_2CH_2Cl	−0.90	
t-$BuCH_2$	−1.74	1.34	Br_2CH	−1.86	0.89
$EtMeCHCH_2$		1.00			
i-$PrCH_2CH_2$	−0.35	0.68	Br_2CMe		1.46
t-$BuCH_2CH_2$	−0.34	0.70	$BrMe_2C$		1.39
t-$BuCHEtCH_2CH_2$		1.01	FCH_2	−0.24	0.62
cyclo-$C_6H_{11}CH_2$	−0.98	0.97	ICH_2	−0.37	0.67
cyclo-$C_6H_{11}CH_2CH_2$		0.70	$CNCH_2$	−0.94	0.89
cyclo-$C_6H_{11}(CH_2)_3$		0.71	I_2CH		0.97
CH_2SMe	−0.34		CHMeBu-*t*	−3.33	2.11
CH_2OPh	−0.33		CMe_2Et	−1.8	
CH_2CH_2OMe	−0.77		*n*-C_8H_{17}	−0.33	0.68
$CH(CH_2CHMe_2)$	−2.47	1.70			
CPh_3	−4.68		CHMeEt	−1.13	1.02
cyclo-C_4H_7	−0.06		$CH_2CH_2CHMe_2$	−0.35	0.68
cyclo-C_5H_9	−0.51		$CH(CH_2Bu\text{-}t)_2$	−3.18	2.03
cyclo-C_6H_{11}	−0.79	0.87	$CHMeCH_2Bu$-*t*	−1.85	1.41
cyclo-C_7H_{13}	−1.10		$CMe(Bu\text{-}t)CH_2Bu$-*t*	−4.00	
i-Pr	−0.47	0.76	F_2CH	−0.67	0.68

[a] Parameters are from O. Exner, in *Correlation Analysis in Chemistry: Recent Advances*, N.B. Chapman and J. Shorter (eds.), Plenum Press, New York, 1978, p. 439. The E_s parameter is sometimes standardised on $E_s(H) = 0$; the figures quoted above refer to E_s standardised on $E_s(Me) = 0$. The two scales are related by a constant increment: $E_s(H) - E_s(Me) = 1.24$. Thus E_s for hydrogen on the Me scale is 1.24.

Table 3 *Polar and steric substituent constants for ortho groups*

Group	E_S	σ_o^*
Methoxy	0.99	−0.22
Ethoxy	0.90	n/a
Fluoro	0.49	0.41
Chloro	0.18	0.37
Bromo	0.00	0.38
Methyl	(0.00)	(0.00)
Iodo	−0.2	0.38
Nitro	−0.75	0.97
Phenyl	−0.90	n/a

n/a = not available
Parameters from M. Charton, *Progr. Phys. Org. Chem.*, 1971, **8**, 235.

Table 4 *Ritchie N_+ parameters*

Nucleophile	*Solvent*	N_+[a]	*Nucleophile*	*Solvent*	N_+[a]
H_2O	water	0.73	Piperidine	water	7.26[b]
NH_3	water	3.89		DMSO	8.68[b]
n-$BuNH_2$	methanol	6.16	OH^-	water	4.75[c]
$EtNH_2$	water	5.28	MeO^-	methanol	7.51
$H_2NCH_2CH_2NH_2$	water	5.44	$HC{\equiv}CCH_2O^-$	water	5.84
$EtO_2CCH_2NH_2$	water	4.40	$CF_3CH_2O^-$	water	5.06
	methanol	5.13	CN^-	water	4.12
	DMSO	6.54		DMF	9.44
$H_2NCH_2CH_2NH_2$	water	4.35		DMSO	8.64
$^-O_2CCH_2NH_2$	water	5.36	N_3^-	water	7.54
Glycylglycinate	water	4.69		methanol	8.78
$MeOCH_2CH_2NH_2$	water	5.07	ClO^-	water	7.41
	DMSO	7.56	HOO^-	water	8.52
n-$PrNH_2$	DMSO	7.88	SO_3^-	water	8.01
$CF_3CH_2NH_2$	water	3.45	BH_4^-	water	6.95
	DMSO	4.86	$C_6H_5S^-$	water	9.10
H_2NNH_2	water	6.01		methanol	10.41
	methanol	6.89	EtS^-	water	8.76
	DMSO	8.17	$HOCH_2CH_2S^-$	water	8.87
$HONH_2$	water	5.05		DMSO	12.71
$MeONH_2$	water	4.37	$MeO_2CCH_2S^-$	water	8.89
	DMSO	6.38		DMSO	12.71
$C_6H_5NHNH_2$	water	4.77	n-PrS^-	water	8.93
$H_2NCONHNH_2$	water	3.73	$^-O_2CCH_2S^-$	water	9.09
	DMSO	6.22			
Morpholine	water	6.78[b]	Morpholine	DMSO	8.56[b]

[a] Based on reactions of malachite green or tri-4-anisylmethyl cation. Data from C.D. Ritchie, *Can. J. Chem.*, 1986, **64**, 2239.
[b] Calculated from N_+ for the pyronin-Y cation.
[c] All values relative to N_+ = 4.75 for HO^- in water.

Table 5 *Swain–Scott, Pearson and Edwards parameters for nucleophilic reagents*

Nucleophile	n_{MeI}	E_n	pK_a	*Nucleophile*	n_{MeI}	E_n	pK_a
MeOH	(0.00)	(0)	−1.7	$(C_2H_5)_3As$	6.90	<2.61	
CO	<2.0			$(C_6H_5)_3P$	7.00	2.73	
MeO^-	6.29	1.65	15.7	$(n\text{-}C_4H_9)_3P$	8.69	8.43	
F^-	2.7	−0.27	3.45	$(C_2H_5)_3P$	8.72		8.69
Cl^-	4.37	1.24	−5.7	$MeCOO^-$	4.3	0.95	4.75
NH_3	5.50	1.36	9.25	$C_6H_5COO^-$	4.5		4.19
Imidazole	4.97		7.10	$C_6H_5O^-$	5.75	1.46	9.89
Piperidine	7.30		11.21	$C_6H_5N(Me)_2$	5.64		5.06
Aniline	5.70	1.78	4.58	2,6-Dimethyl-pyridine	3.51		
Pyridine	5.23	1.20	5.23	2-Picoline	4.7		6.48
NO_2^-	5.35	1.73	3.37	Pyrrolidine	7.23		11.27
$(C_6H_5CH_2)_2S$	4.84			*N,N*-Dimethyl-cyclo-hexylamine	6.73		
N_3^-	5.78	1.58	4.74	$(C_2H_5)_3N$	6.66		10.70
NH_2OH	6.60		5.82	$(C_2H_5)_2NH$	7.0		11.0
NH_2NH_2	6.61		7.93	$C_6H_5Se^-$	10.7		
C_6H_5SH	5.70			$(C_6H_5)_3SiO^-$	6.2		
Br^-	5.79	1.51	−7.7	$C_6H_5SO_2NCl^-$	6.8		3.0
$(C_2H_5)_2S$	5.34			$C_6H_5SO_2NH^-$	5.1		8.5
$((C_2H_5)_2N)_3P$	8.54			Phthalimide anion	5.4		7.4
$(Me)_2S$	5.54		−5.3	HS^-	8	2.10	7.8
$(MeO)_2PO$	7.00			SO_4^{2-}	3.5	0.59	2.0
$(CH_2)_5S$	5.42			NO_3^-	1.5	0.29	−1.3
$(CH_2)_4S$	5.66		−4.8	$(C_6H_5)_3Ge^-$	12		
$SnCl_3^-$	3.84			$(C_6H_5)_3Sn^-$	11.5		
I^-	7.42	2.06	−10.7	$(C_6H_5)_3Pb^-$	8		
$(C_6H_5CH_2)_2Se$	5.23			$Re(CO)_5^-$	8		
$(Me)_2Se$	6.32			$Mn(CO)_5^-$	5.5		
SCN^-	6.70	1.83	−0.7	$Co(CO)_4^-$	3.5		
SO_3^{2-}	8.53	2.57	7.26	HO_2^-	7.8		
C_6H_5NC	<2.0			$C_6H_5S^-$	9.92	2.9	6.52
$(C_6H_5)_3Sb$	<2.0			$SC(NH_2)_2$	7.27	2.18	−0.96
$(C_6H_5)_3As$	4.77			$(MeO)_3P$	5.2		
$SeCN^-$	7.85			$S_2O_3^{2-}$	8.95	2.52	1.9
CN^-	6.70	2.79	9.3				

Data from R.G. Pearson, *J. Am. Chem. Soc.*, 1968, **90**, 319.

Table 6 *Y values of solvent ionising power for some solvents and solvent mixtures (with water) derived from solvolyses of 1- or 2-adamantyl tosylates at 25°C*

Composition/%	*Solvent*	$Y_{G\text{-}W}$ [a]	Y_{OTs} [b]	N_{OTs} [b]
[illegible]	EtOH	−2.03	−1.96	0.06
[illegible]	EtOH	−0.75	−0.77	0.07
80	EtOH	0.00	0.00	0.00
70	EtOH	0.60	0.47	−0.05
60	EtOH	1.12	0.92	−0.08
50	EtOH	1.66	1.29	−0.09
40	EtOH		1.97	−0.23
30	EtOH		2.84	−0.35
20	EtOH		3.32	−0.34
10	EtOH		3.78	−0.41
	water	3.49	4.1	−0.44
100	MeOH	−1.09	−0.92	−0.04
90	MeOH	−1.09	−0.05	−0.05
80	MeOH	0.38	0.47	−0.05
70	MeOH	0.96	1.02	−0.08
60	MeOH	1.49	1.52	−0.13
50	MeOH	1.97	2.00	−0.19
40	MeOH		2.43	−0.21
30	MeOH		2.97	−0.31
20	MeOH		3.39	−0.35
10	MeOH		3.78	−0.41
95	acetone		−2.95	
90	acetone	−1.86	−1.99	−0.39
80	acetone	−0.67	−0.94	−0.42
70	acetone	0.13	0.07	−0.42
60	acetone	0.80	0.66	−0.41
50	acetone	1.40	1.26	−0.39
40	acetone		1.85	−0.38
30	acetone		2.50	−0.40
20	acetone		3.05	−0.38
10	acetone		3.58	−0.41

Composition/%	*Solvent*	$Y_{G\text{-}W}$ [a]	Y_{OTs} [b]	N_{OTs} [b]
90	dioxan		−2.41	−0.51
80	dioxan		−1.30	−0.29
100	MeCN		−3.21	
50	MeCN		1.2	
35	MeCN		1.8	
30	MeCN		1.9	
25	MeCN		2.5	
20	MeCN		2.7	
10	MeCN		3.6	
100	$(Me)_2CHOH$	−2.73	−2.83	0.12
100	$(Me)_3COH$		−3.74	
100	CF_3CH_2OH		1.77	−3.07
97	CF_3CH_2OH		1.83	−2.79
85	CF_3CH_2OH		1.92	−2.01
70	CF_3CH_2OH		2.00	−1.20
50	CF_3CH_2OH		2.14	−0.93
100	$(CF_3)_2CHOH$(HFIP)		3.82	
97				
90				
100	$MeCO_2H$(A)	−1.64	−0.90	−2.28
75	A/F		0.70	
50	A/F		1.62	
25	A/F		2.34	
100	HCO_2H(F)	2.05	3.04	−2.35
100	CF_3CO_2H(TFA)	1.84	4.57	−5.56
80	TFA/EtOH	0.41	0.98	−1.72
60	TFA/EtOH		0.21	−1.01
40	TFA/EtOH		−0.44	−0.55
60	H_2SO_4/H_2O		5.29	−0.71
40	H_2SO_4/H_2O		4.67	−1.20
20	H_2SO_4/H_2O		4.39	−2.02
100	HCONHMe		−1.70	
100	MeCONHMe		−2.69	
100	$HCON(Me)_2$		−4.14	
100	$MeCON(Me)_2$		−4.99	

Table 7 *Some Leo–Hansch π parameters for use in calculating logP values*

Substituent	*$ArOCH_2CO_2H$*	*ArOH*	*Substituent*	*$ArOCH_2CO_2H$*	*ArOH*
H	0	0	3-COOH	–0.15	0.04
2-F	0.01	0.25	4-COOH		0.12
3-F	0.13	0.47	2-OMe	–0.33	
4-F	0.15	0.31	3-OMe	0.12	0.12
2-CI	0.59	0.69	4-OMe	–0.04	–0.12
3-CI	0.76	1.04	$3\text{-}OCF_3$	1.21	
4-CI	0.70	0.93	$3\text{-}OCH_2COOH$		–0.70
2-COMe	0.01		$4\text{-}OCH_2COOH$		–0.10
3-COMe	–0.28	–0.07	$3\text{-}NME_2$		0.10
4-COMe	–0.37	–0.11	$3\text{-}NH_2$		–1.29
3-CN	–0.30	–0.24	$4\text{-}NH_2$		–1.63
4-CN	0.32	0.14	$2\text{-}NO_2$	–0.23	0.33
2-Br	0.75	0.89	$3\text{-}NO_2$	0.11	0.54
3-Br	0.94	1.17	$4\text{-}NO_2$	0.24	0.50
4-Br	1.02	1.13	3-NHCOPh	–0.72	
2-I	0.92	1.19	$3\text{-}NHCONH_2$	–1.01	
3-I	1.15	1.47	3-NHCOMe	–0.79	
4-I	1.26	1.45	3-*n*Bu	1.90	
2-Me	0.68		3-*t*Bu	1.68	
3-Me	0.51	0.56	4-*Cyclo*hexyl	2.51	
4-Me	0.52	0.48	$3,4[CH_2]_3$	1.04	
2-Et	1.22		$3,4[CH]_4$	1.24	
3-Et	0.97	0.94	$3,4[CH_2]_4t$	1.39	
3-*n*Pr	1.43		3-OH	–0.49	–0.66
3-*i*Pr	1.30		4-OH	–0.61	–0.87
4-*i*Pr	1.40				
4-*sec*Bu	1.82		4N = NPh	1.71	
4-*Cyclo*pent	2.14		3-SMe	0.62	
3-Ph	1.89		$3\text{-}SO_2CF_3$	0.93	
			$3\text{-}CH_2\text{-}COOH$		–0.61
$3\text{-}CF_3$	1.07	1.49	$3\text{-}CH_2OH$		–1.02
$3\text{-}SCF_3$	1.58		$4\text{-}CH_2OH$		–1.26

Data taken from: T. Fujita *et al.*, *J. Am. Chem. Soc.*, 1964, **86**, 5175.

Table 8 *Fragmental constants, f, for calculating logP*

Calculation step	*f*	*Calculation step*	*f*
CH_2	0.66	Single bond in chain	$-0.12(f_b)$
H	0.23	CH	0.65
Single carbon (C)	0.20	Branching in C chain	$-0.13(f_{cbr})$
Single bond in ring	$-0.09(f_b)$	Branching at a group	$-0.22\ (f_{gbr})$
Phenyl group	1.90	Me	0.89
Aliphatic groups:			
COOH	–1.09	NH_2	–1.54
CN	–1.28	$CONH_2$	–2.18
CI	0.06	I	0.60
NH	–2.11	OH	–1.64
COR_2	–1.90	$-O^-$	–1.81
F	–0.38	Br	0.20
NO_2	–1.26	$-S^-$	–0.79
Aromatic groups:			
COOH	–0.03	NH_2	–1.00
CN	–0.34	$CONH_2$	–1.26
CI	0.94	I	1.35
NH	–1.03	OH	–0.40
CO	–0.32	$-O^-$	–0.57
F	0.37	Br	1.09
NR_2	–1.17	$-CONR_2$	–2.82
–CO–OR	–0.56	CONHR	–1.81
NO_2	–0.02	$-S^-$	0.03

Data taken from A.J. Leo *et al.*, *J. Med. Chem.*, 1975, **18**, 865.
R. Rekker and R. Mannhold, *Calculations of Drug Lipophilicity*, VCH, Weinheim, 1992; H. Kubinyi, *Prog. Drug Res.*, 1979, **23**, 97; C. Hansch and A.J. Leo, *Substituent Constants for Correlation Analysis in Chemistry and Biology*, Wiley-Interscience, New York, 1979, pp. 18–43.